

LambdaEta yield from production-vertex missing mass

Macro: plot_lambda_eta_yield.cc | Executed: 2026-10-01 15:07:29

Beam momentum setting: 715 MeV/c

Effective nTPC = 2

Runs used: 02682, 02683, 02684, 02686, 02687, 02689, 02690, 02691, 02692, 02693

Input: /gpfs/home/had/haein/data/JPARC2025Nov_root/physics-715/runXXXXXX_DstTPCLambdaEta.root

Selection: X_prod_found; Lambda mass window $1.09 < M(p \pi^-) < 1.13 \text{ GeV}/c^2$

Lambda mass displayed: 1.04 to 1.20 GeV/c^2 ; peak fit: 1.10 to 1.13 GeV/c^2

Pre-cut Lambda peak fit model: Gaussian + linear background, 1.100 to 1.130 GeV/c^2

Production vertex LH2 cut: on

Pion selection: lambda_pion_track_id matched to pion_beam_window_status == 0 when branch exists

Combined eta fit: Gaussian + linear background up to kinematic endpoint, 0.50 to 0.60 GeV/c^2

Linear background fit: 0.00--0.53 GeV/c^2 ; fixed in the eta-peak fit

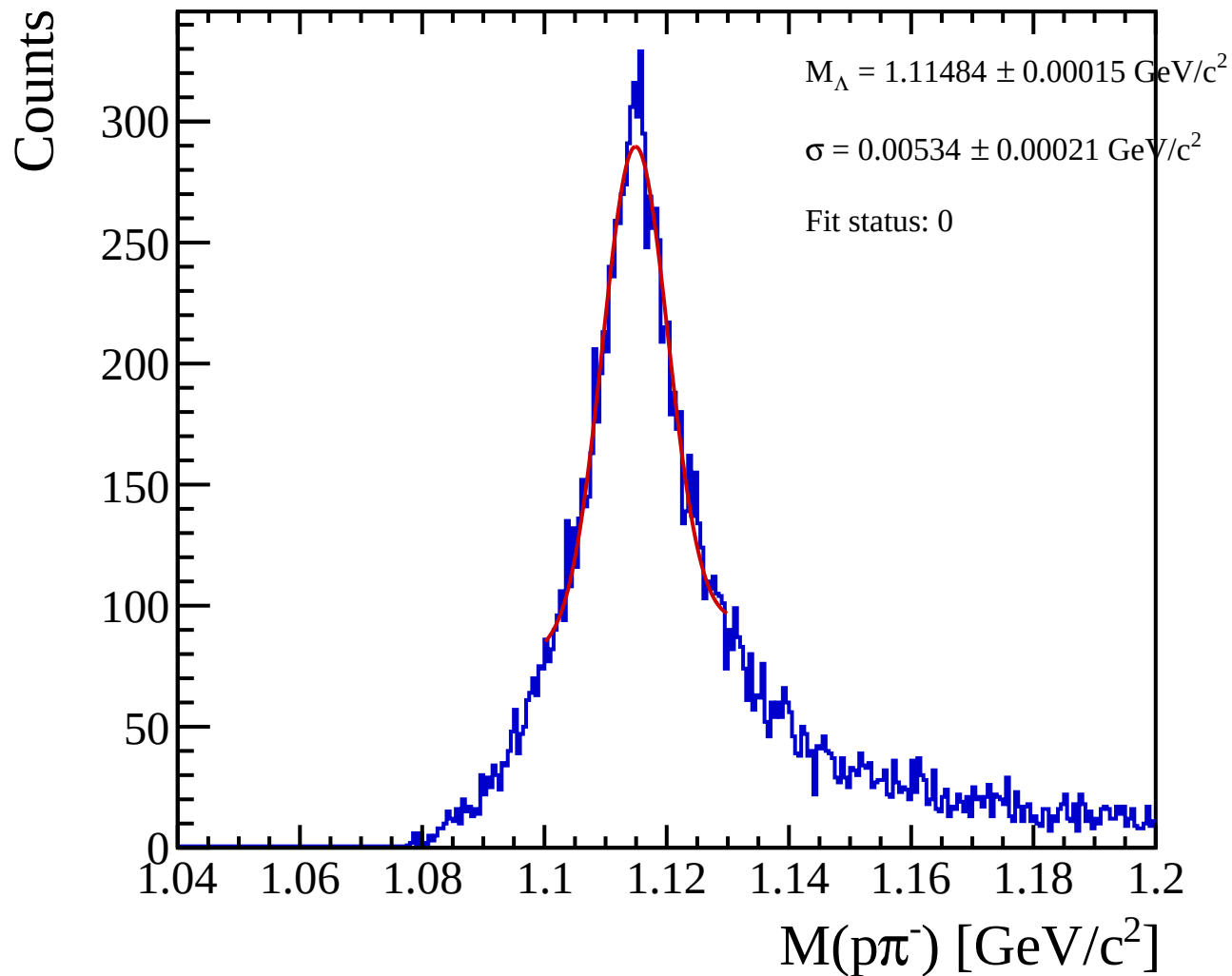
Selected-candidate mean beam momentum = 0.7319 GeV/c ; kinematic endpoint = 0.5519 GeV/c^2

Momentum-dependent eta yield: Gaussian signal area from the combined fit

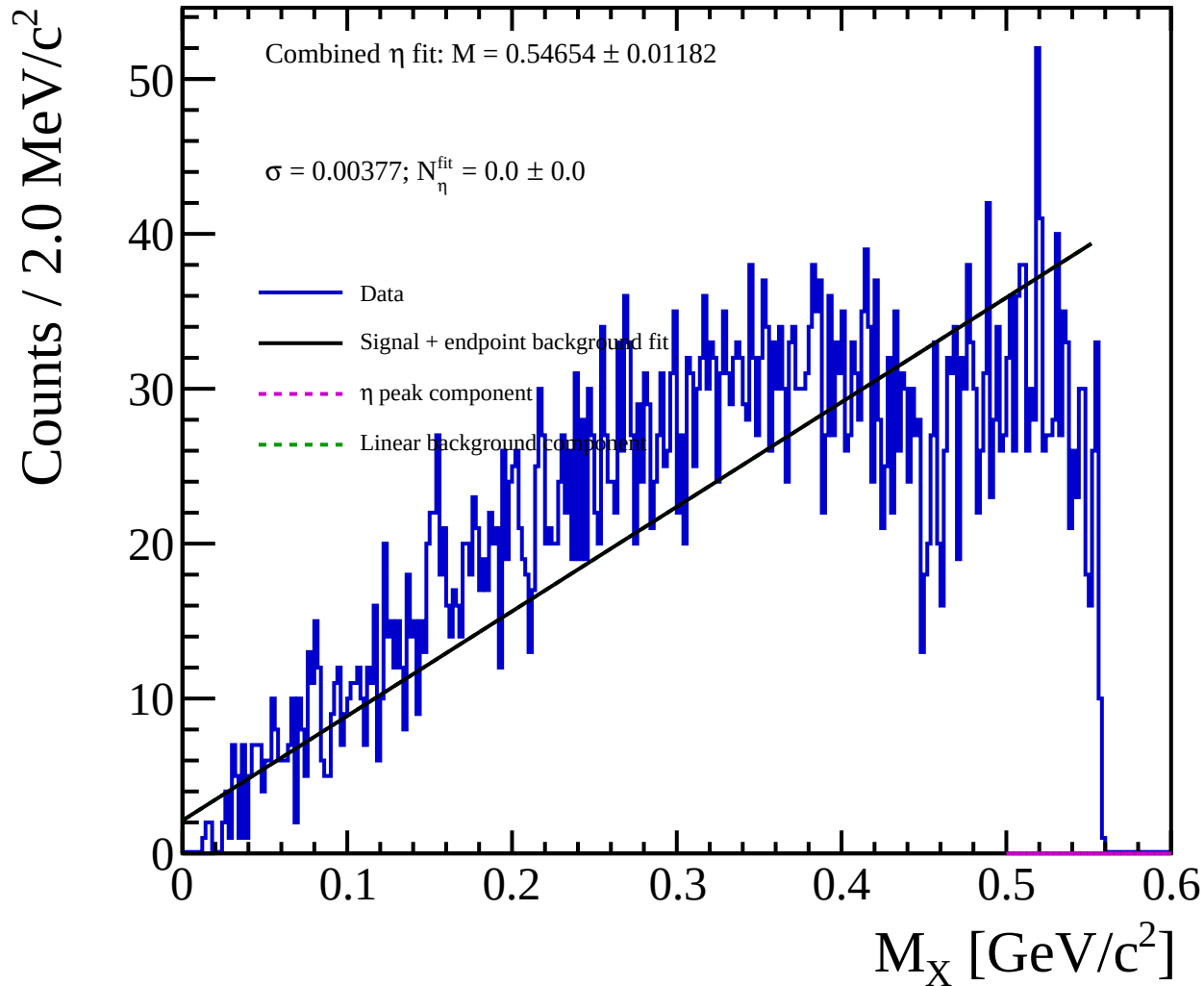
Production beam momentum: 25 bins x 4 MeV/c from 700 to 800 MeV/c

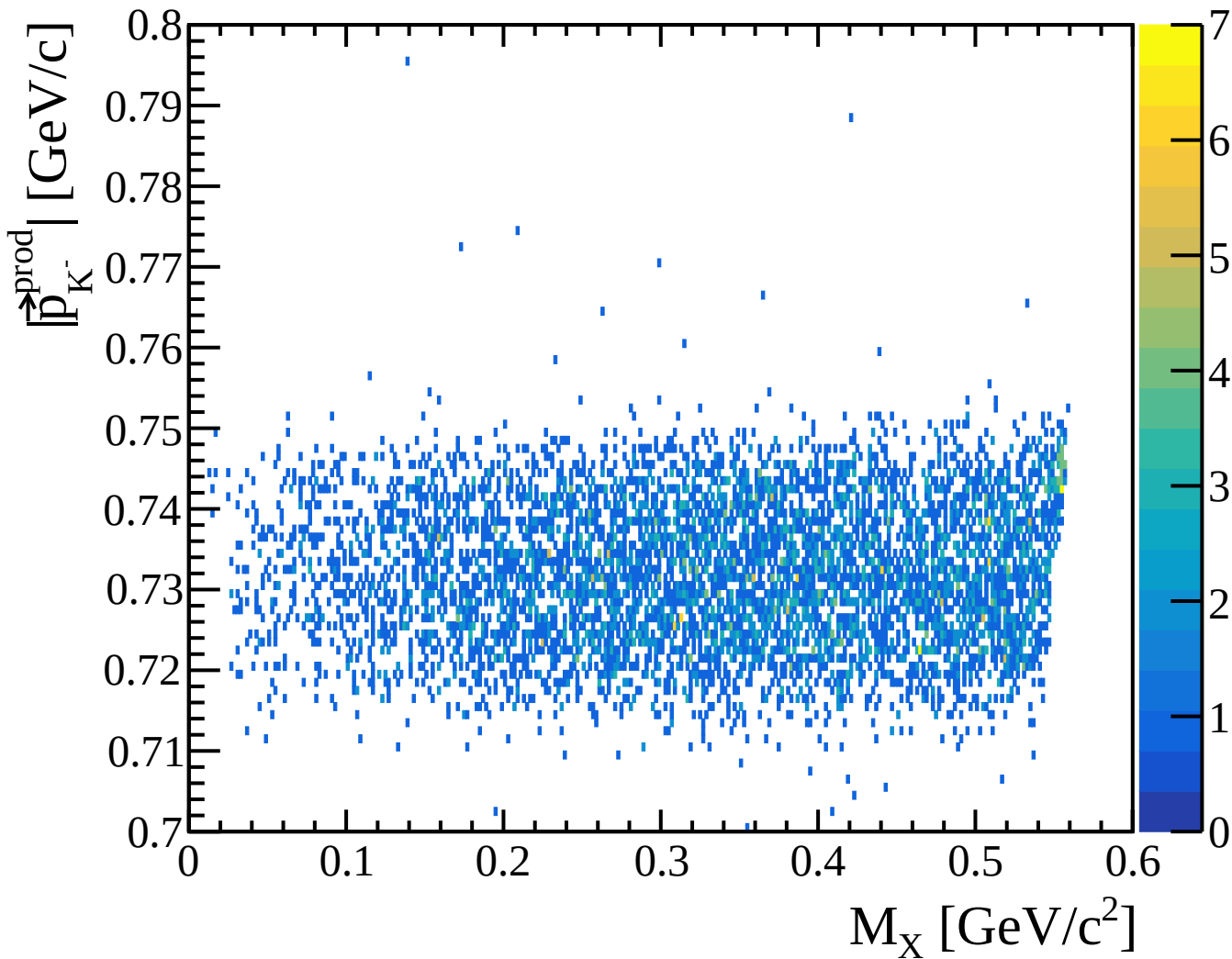
Events read: 1716788 | candidates checked: 54083 | pre-cut Lambda entries: 20713 | eta selection: 11602

Λ invariant mass before Lambda-mass cut

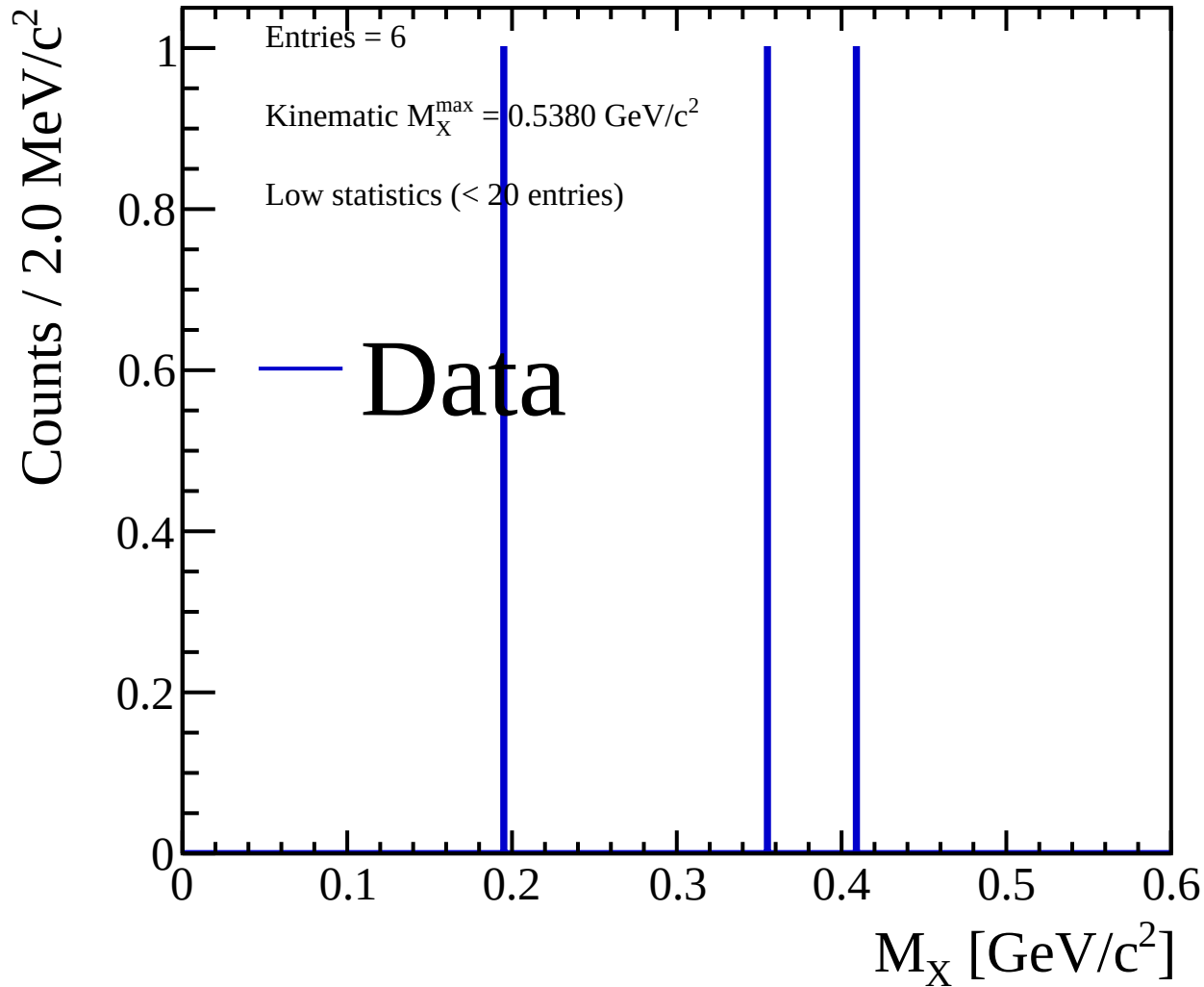


LambdaEta missing mass

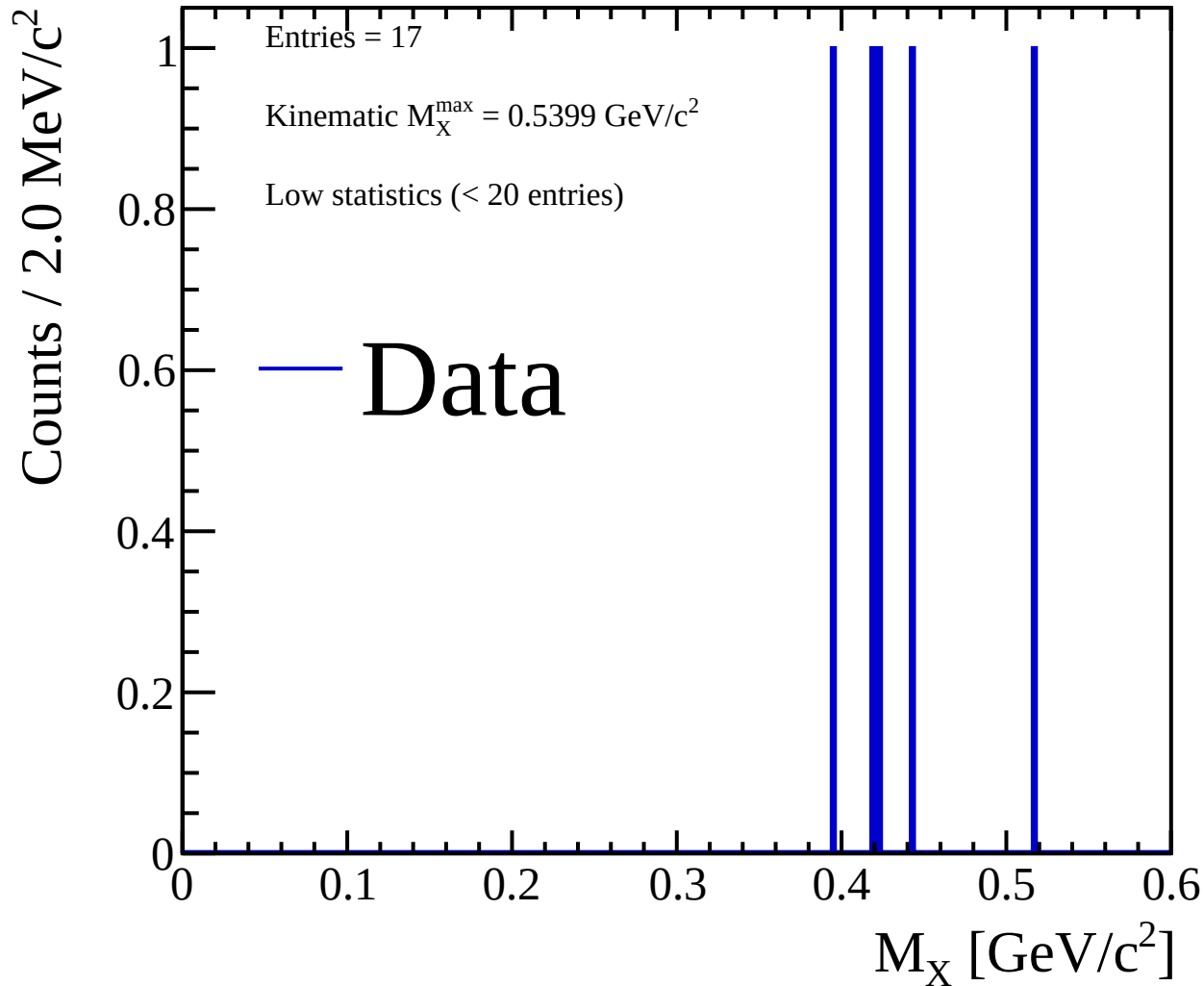


Production-vertex K^- momentum vs $\Lambda\eta$ missing mass

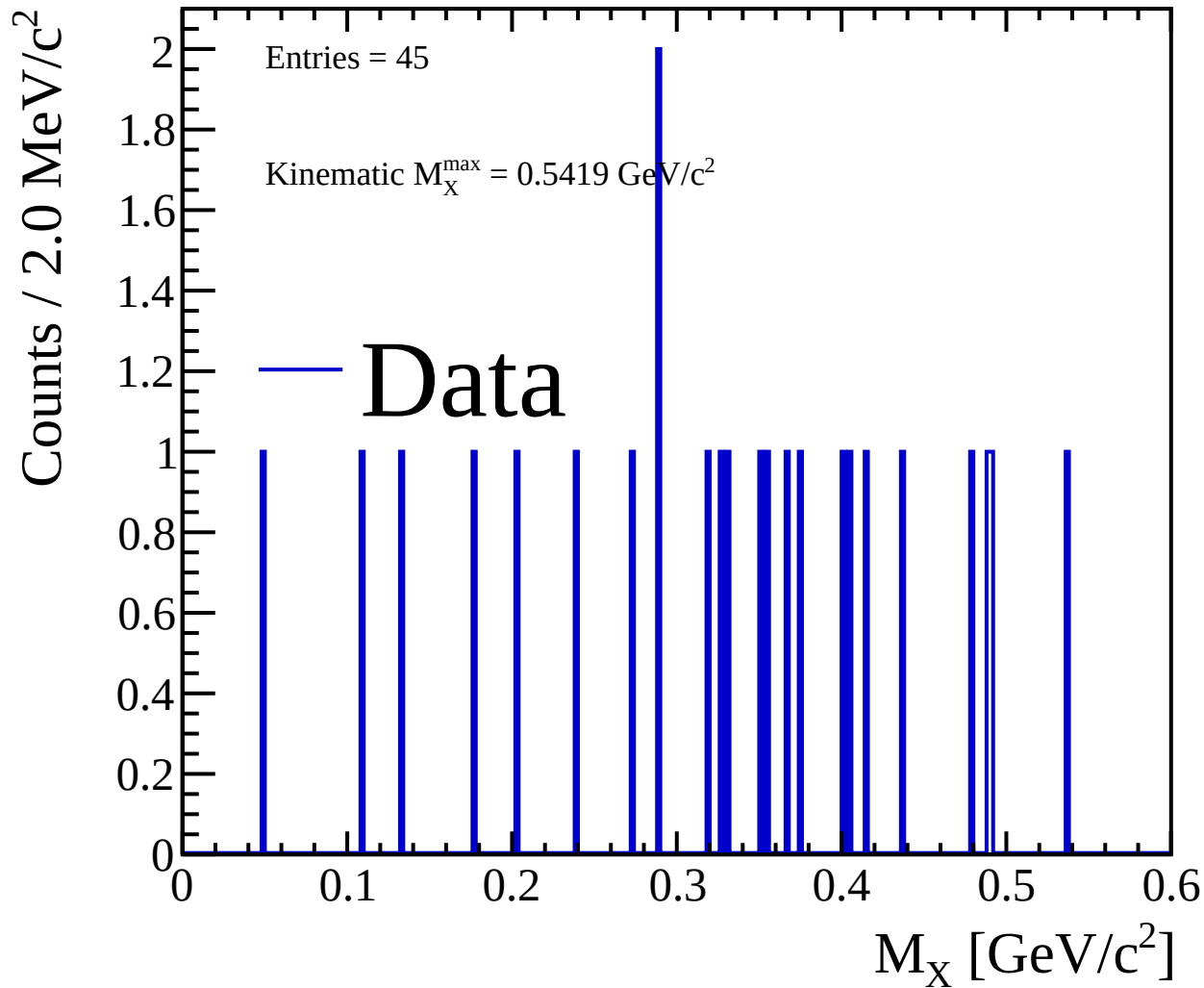
Production-vertex K^- momentum 700--704 MeV/c



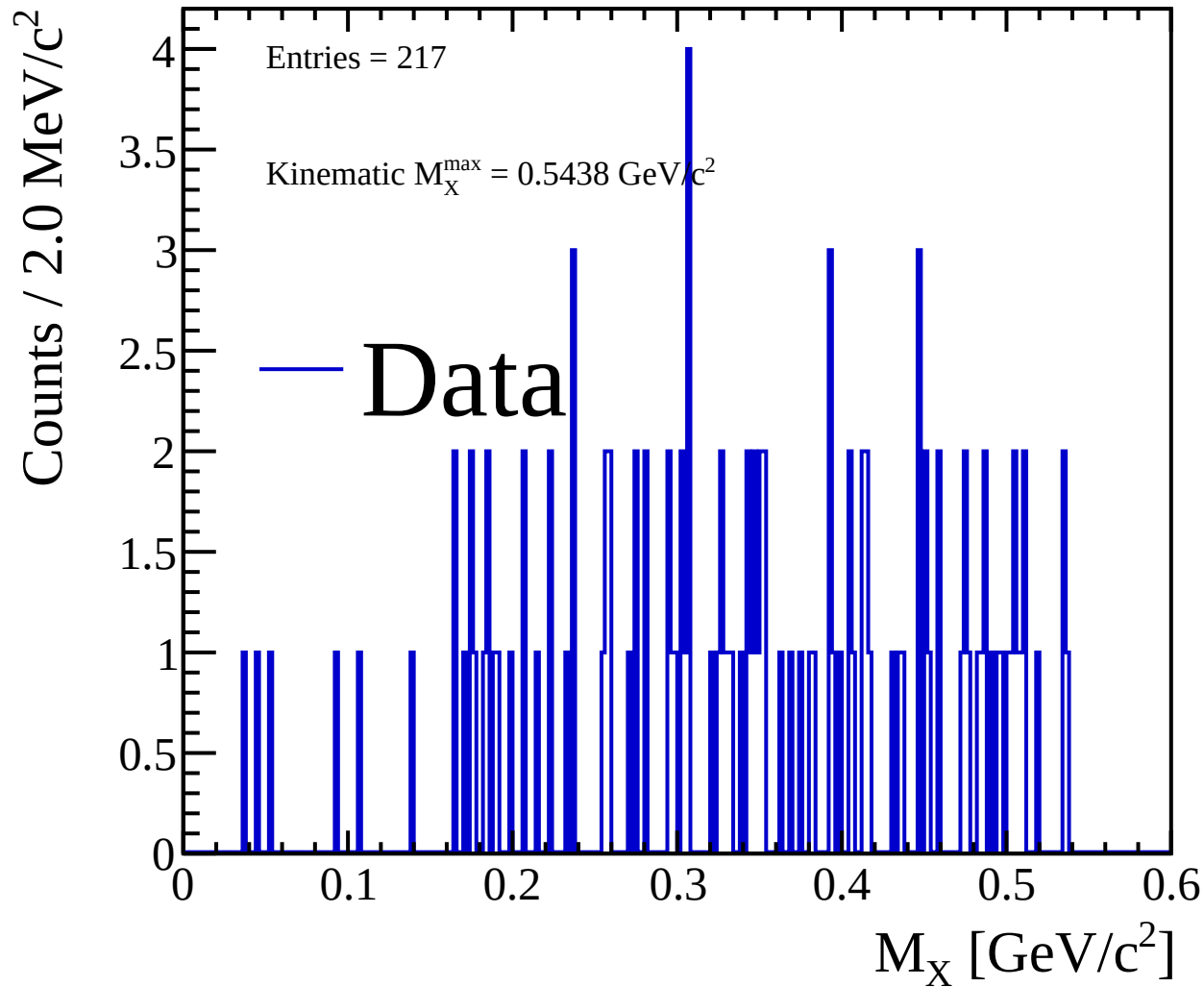
Production-vertex K^- momentum 704--708 MeV/c



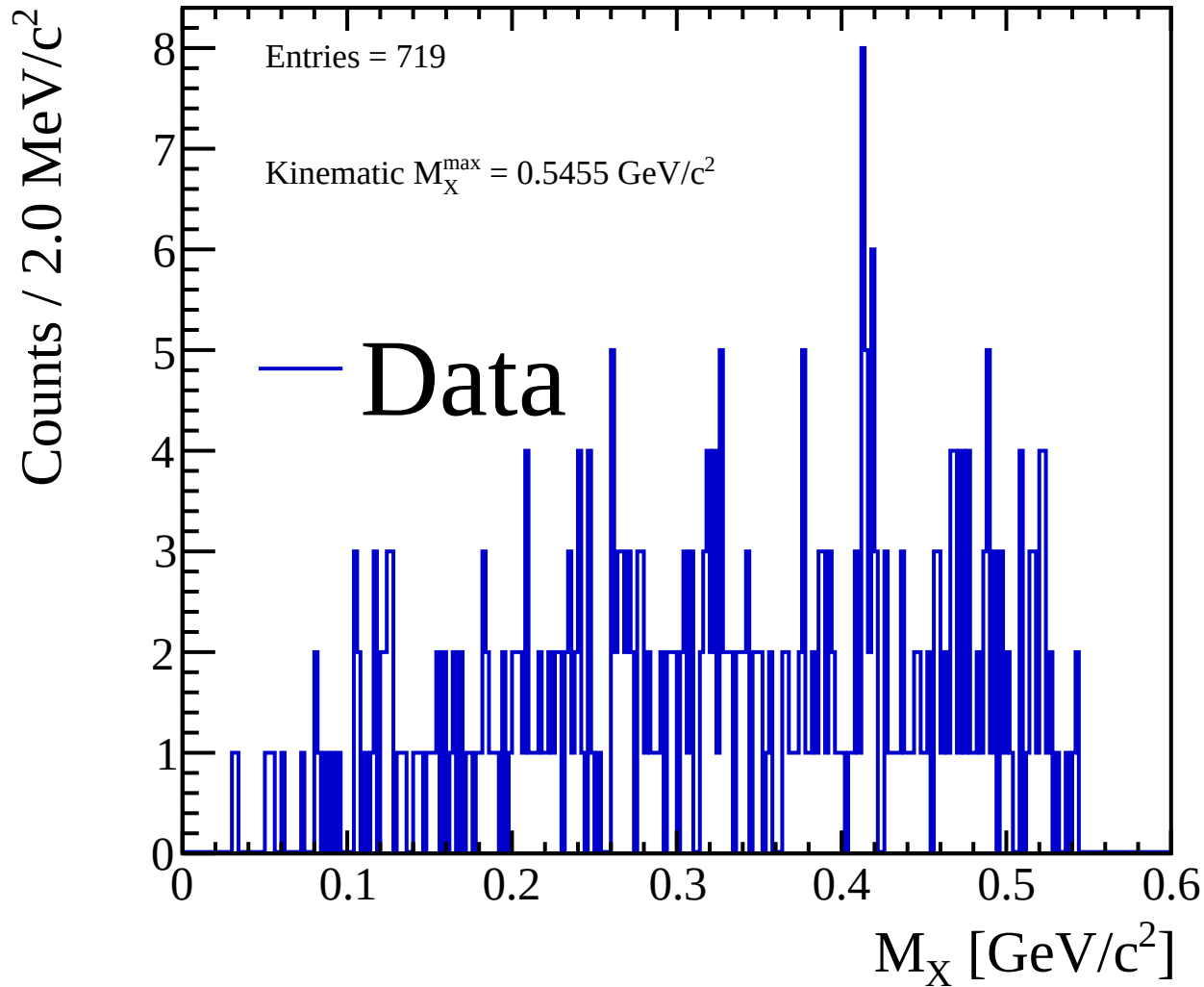
Production-vertex K^- momentum 708--712 MeV/c



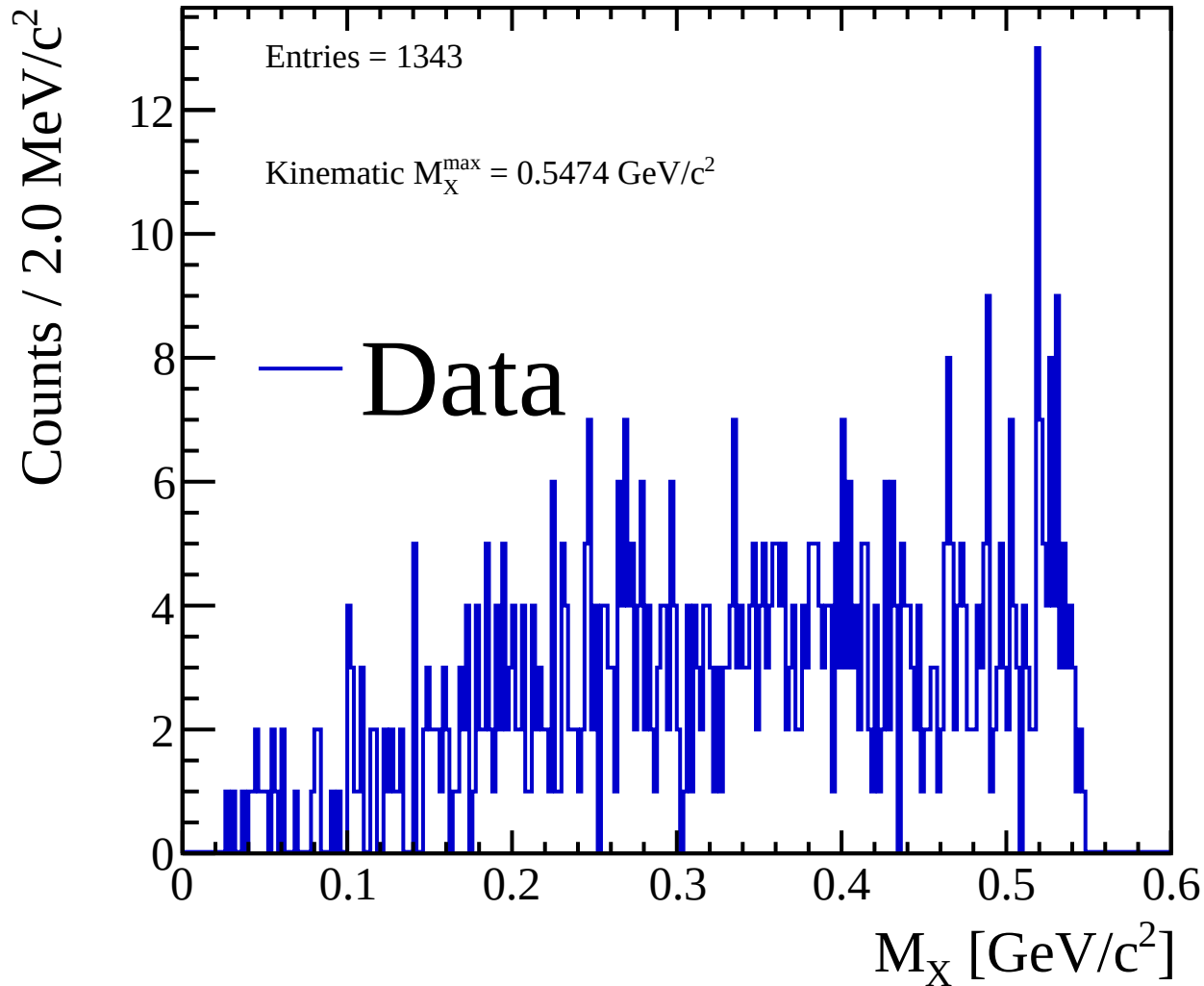
Production-vertex K^- momentum 712--716 MeV/c



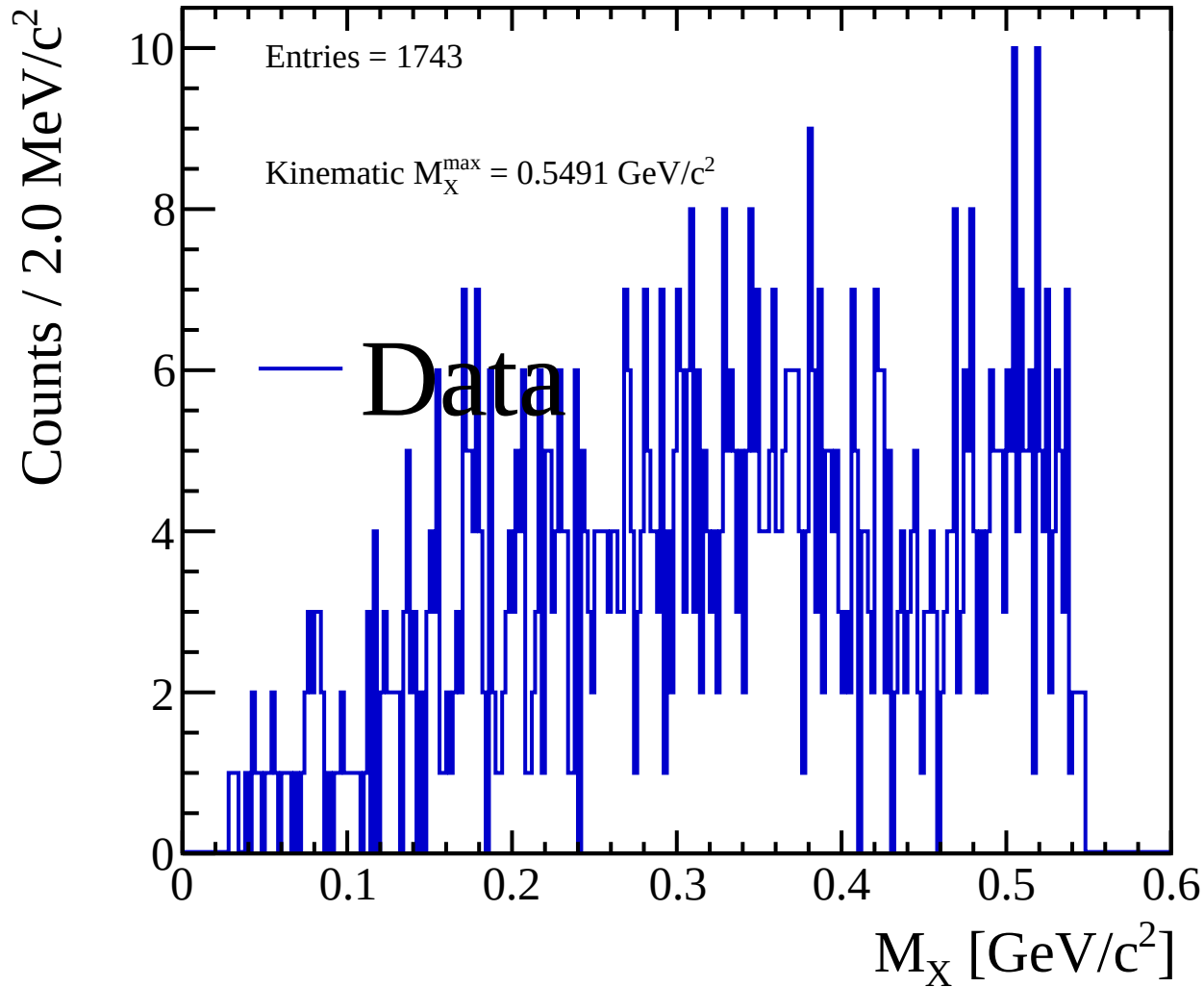
Production-vertex K^- momentum 716--720 MeV/c



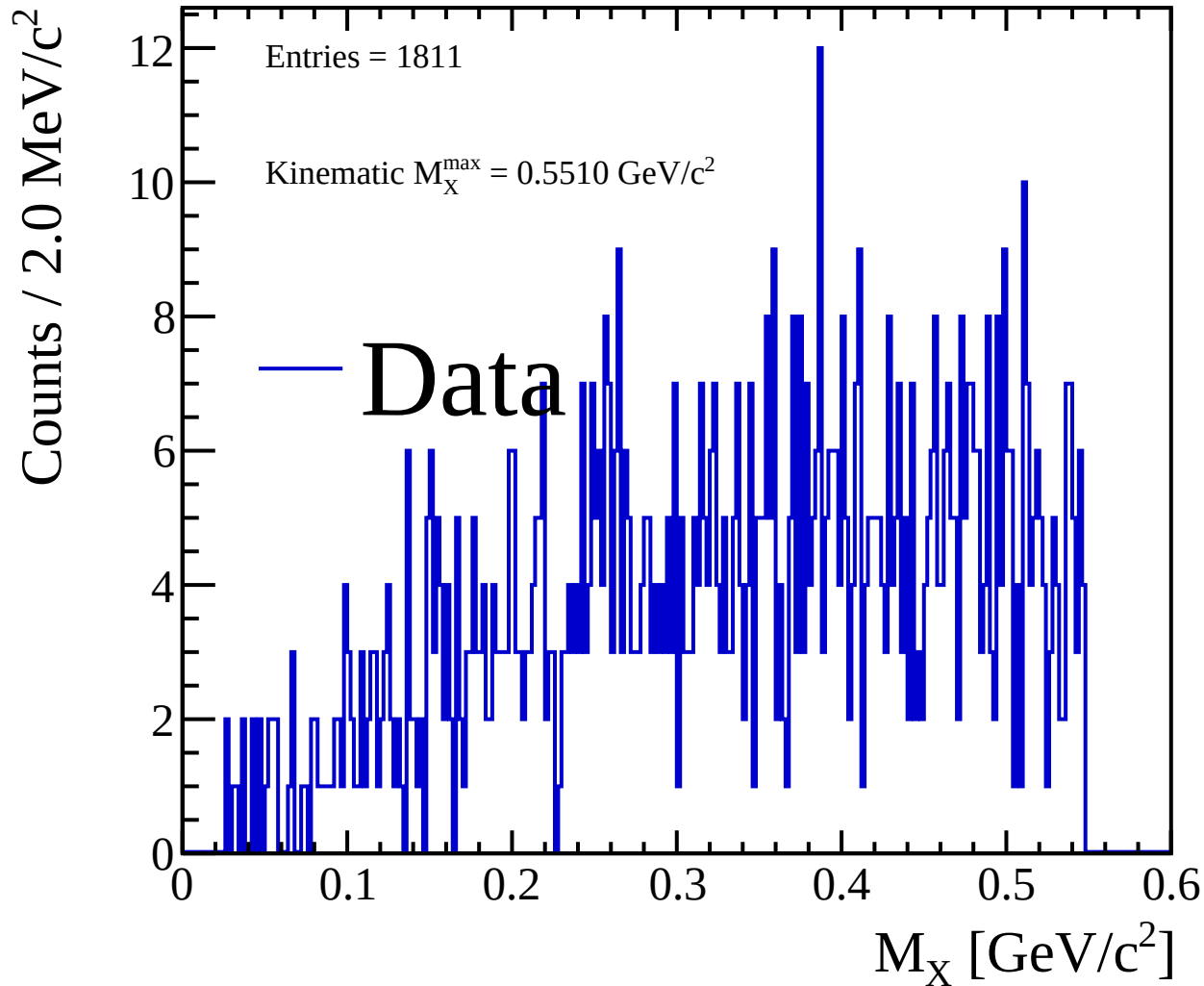
Production-vertex K^- momentum 720--724 MeV/c



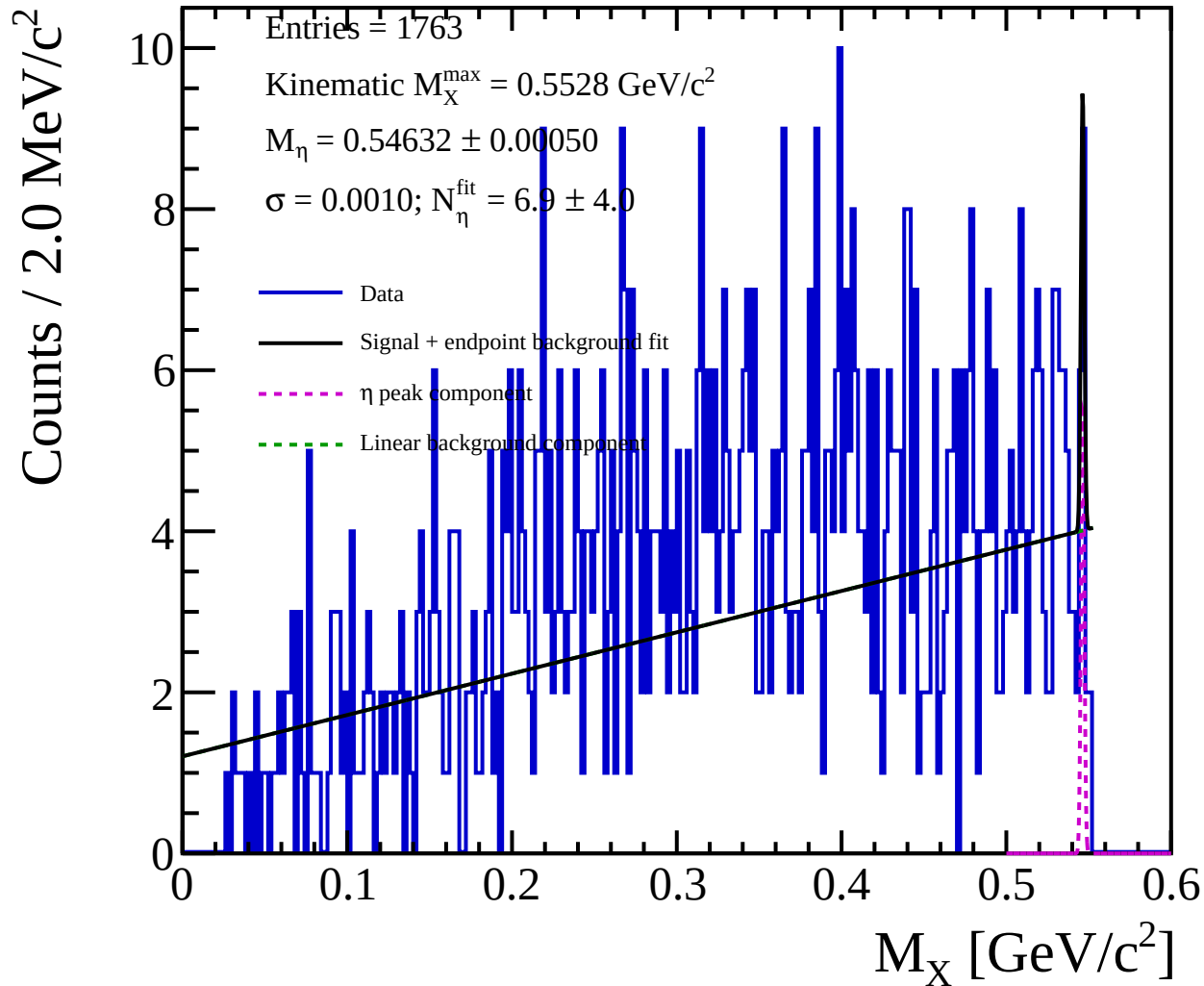
Production-vertex K^- momentum 724--728 MeV/c



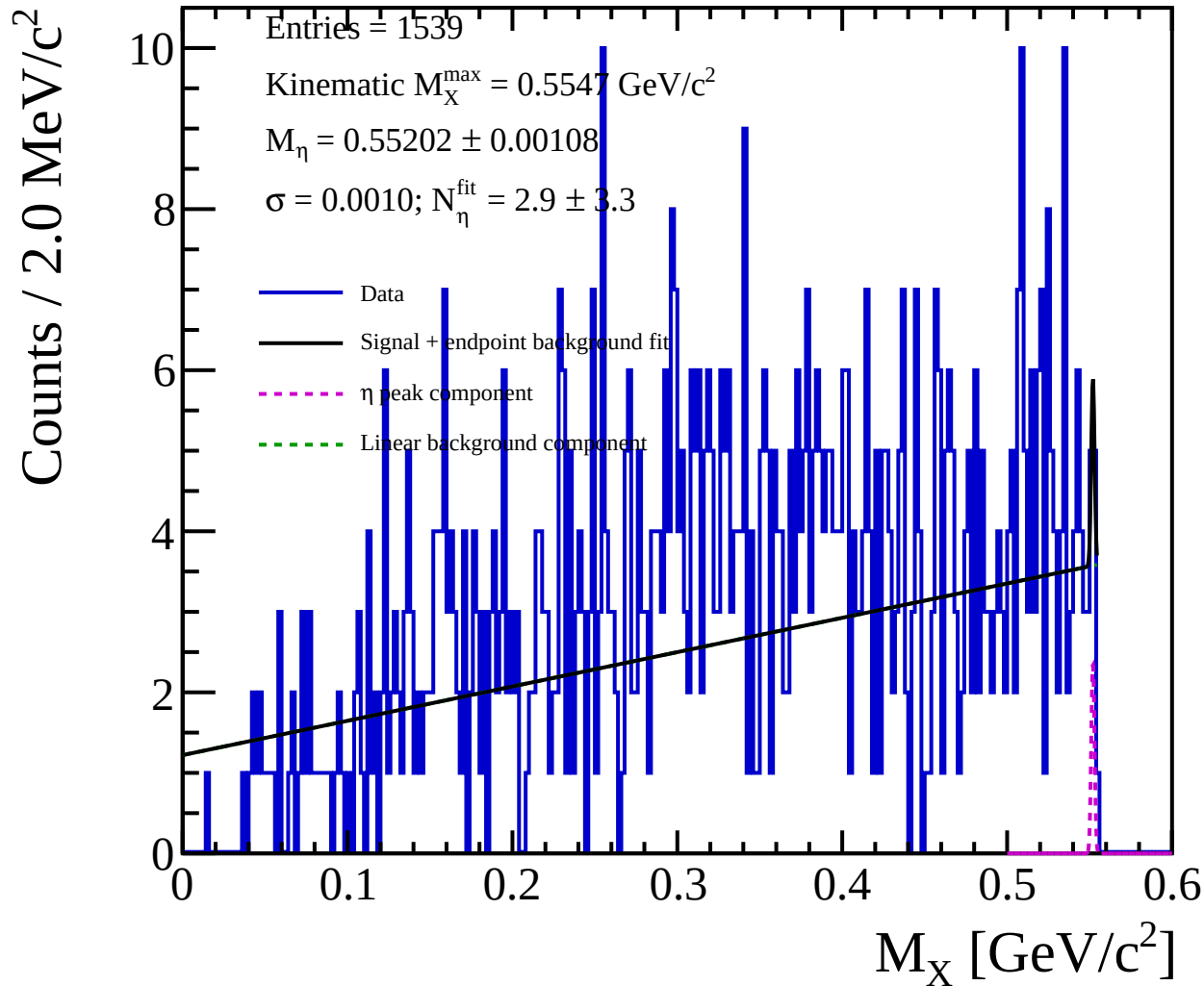
Production-vertex K^- momentum 728--732 MeV/c



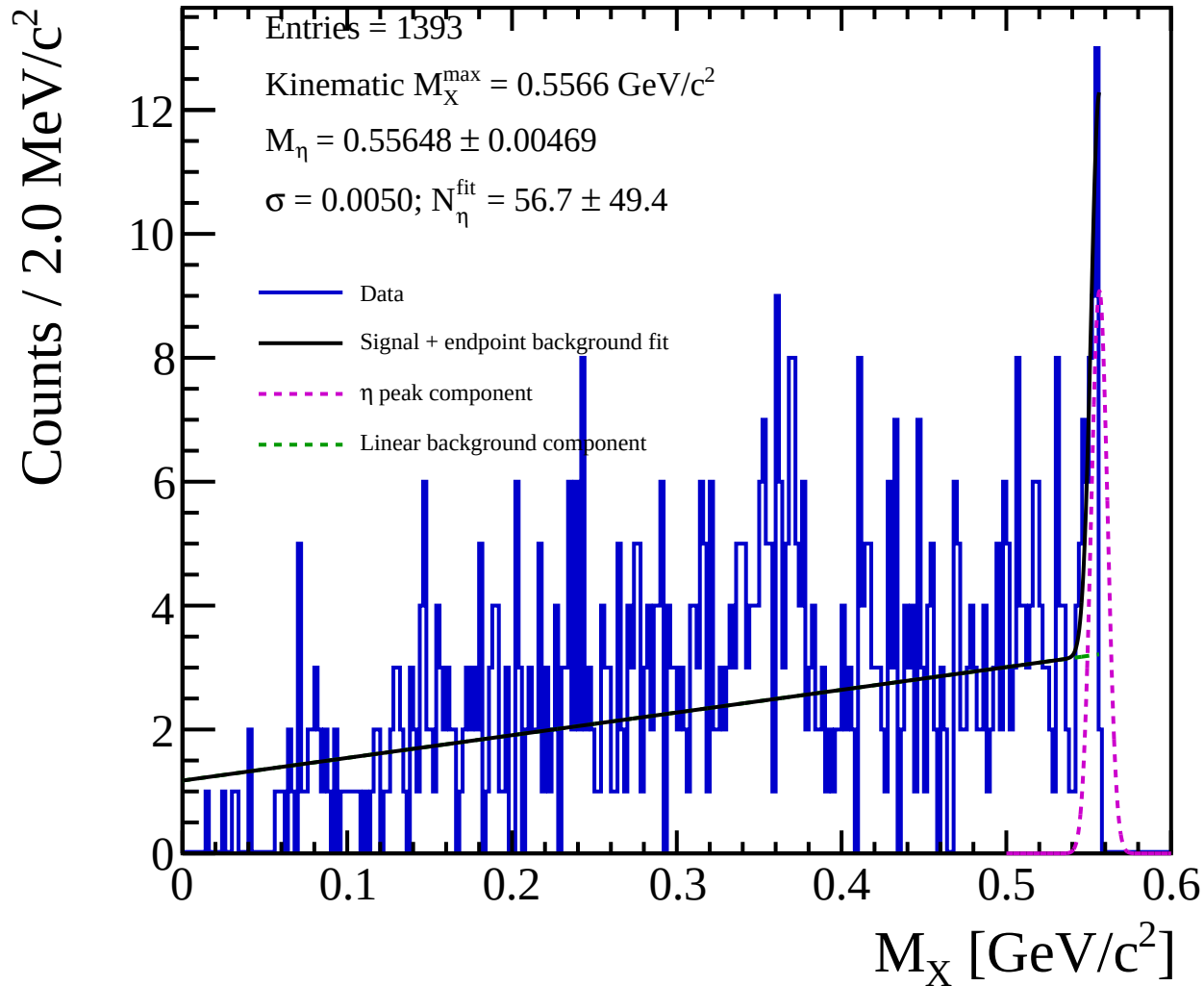
Production-vertex K^- momentum 732--736 MeV/c



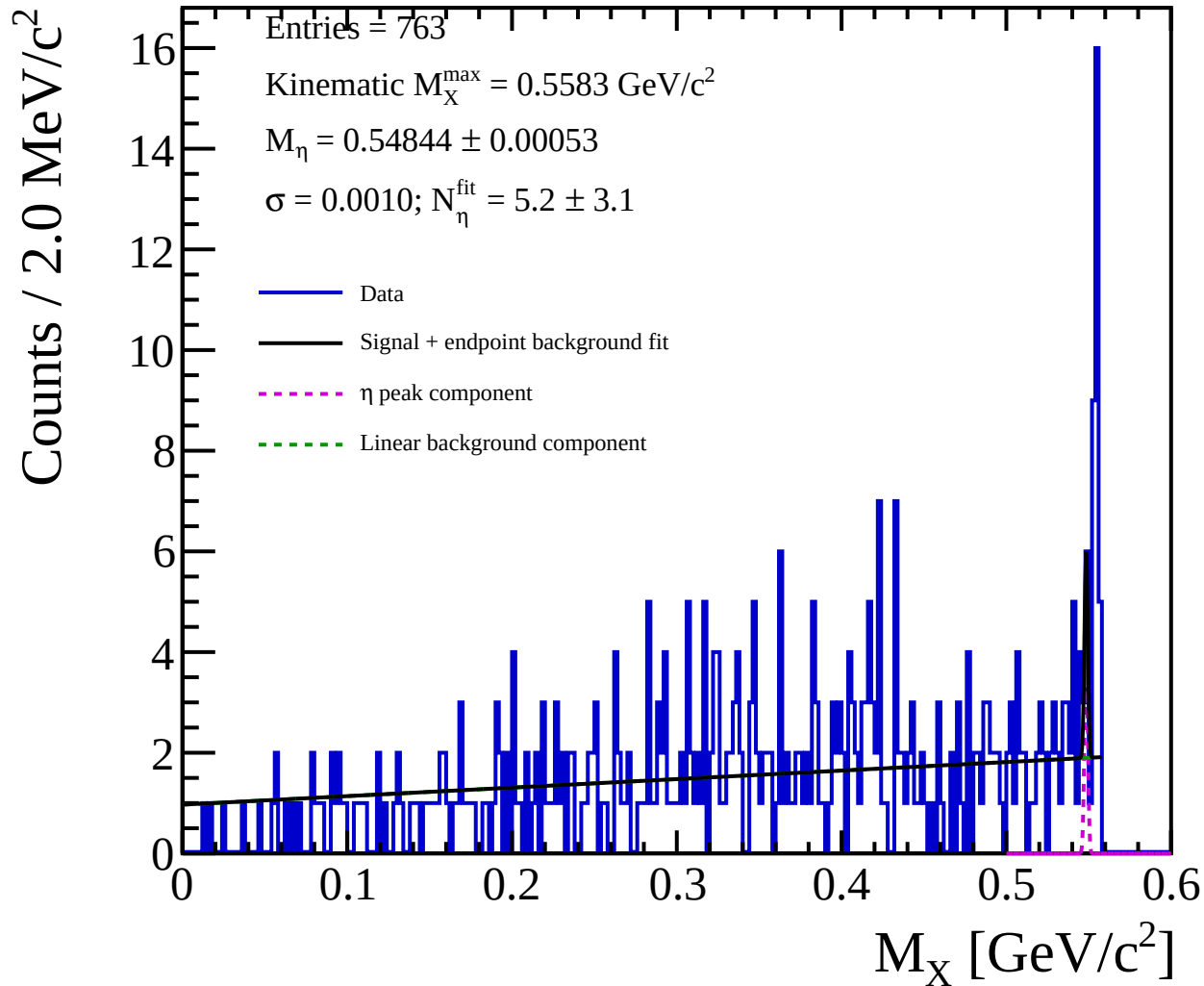
Production-vertex K^- momentum 736--740 MeV/c



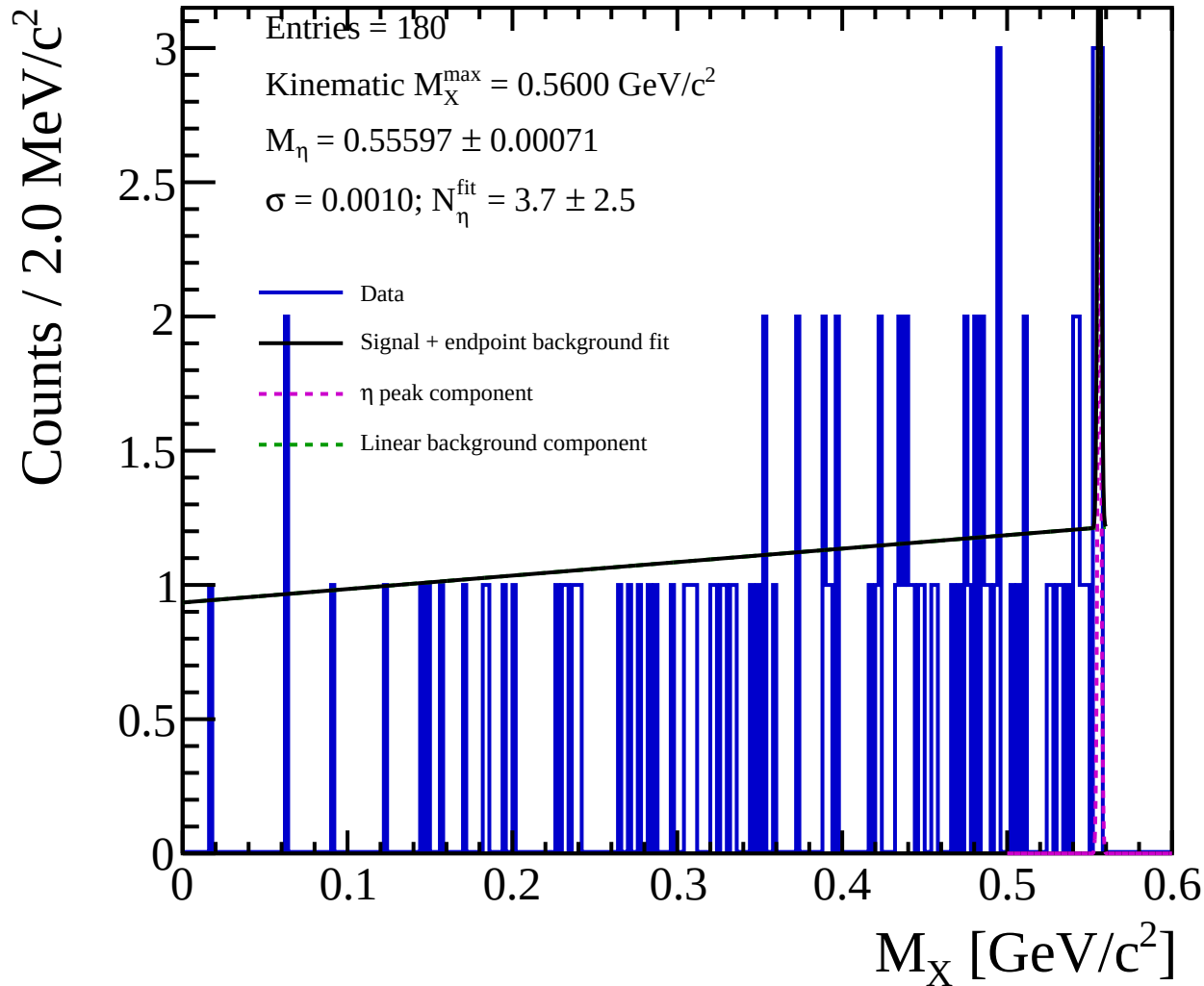
Production-vertex K^- momentum 740--744 MeV/c



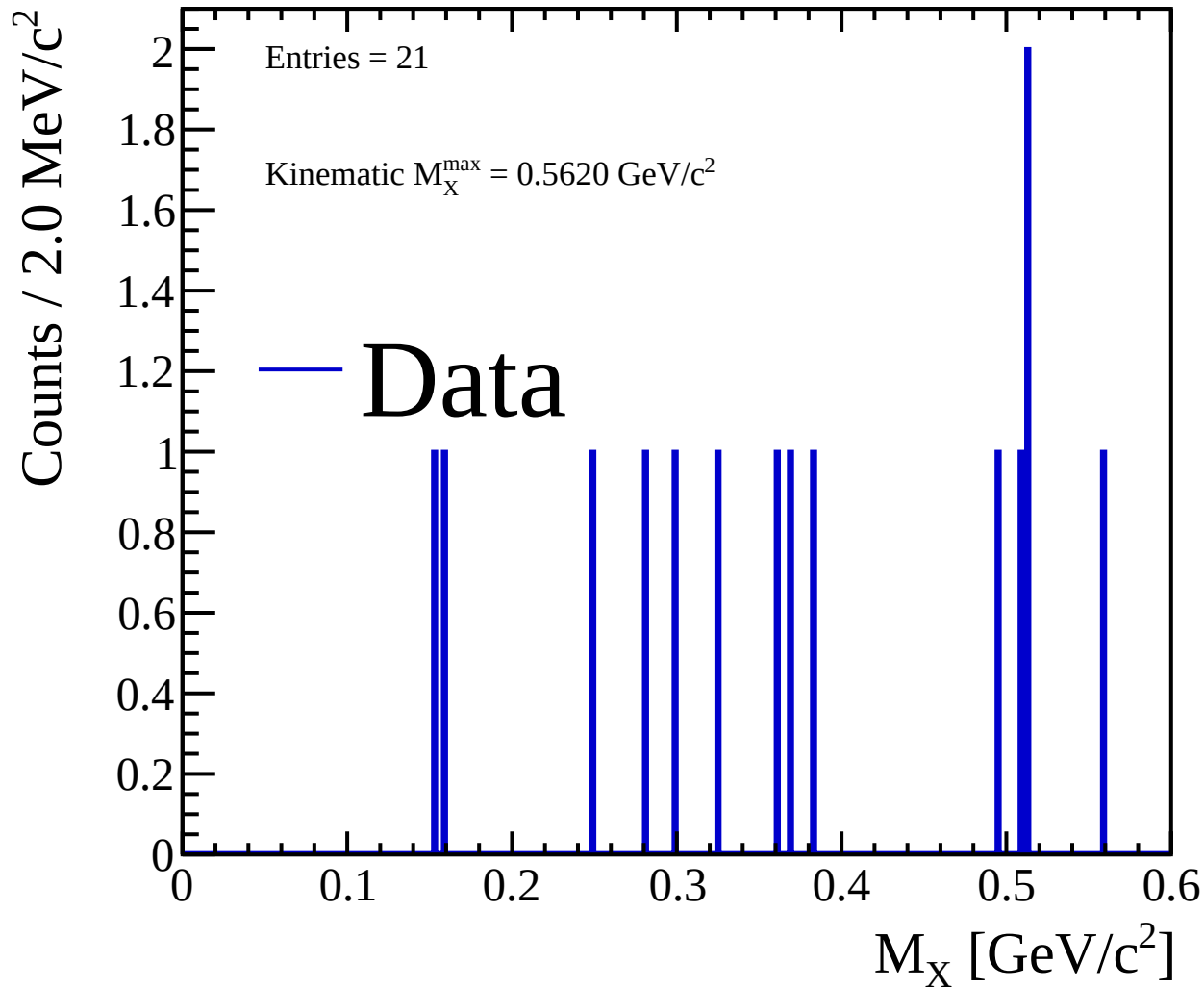
Production-vertex K^- momentum 744--748 MeV/c



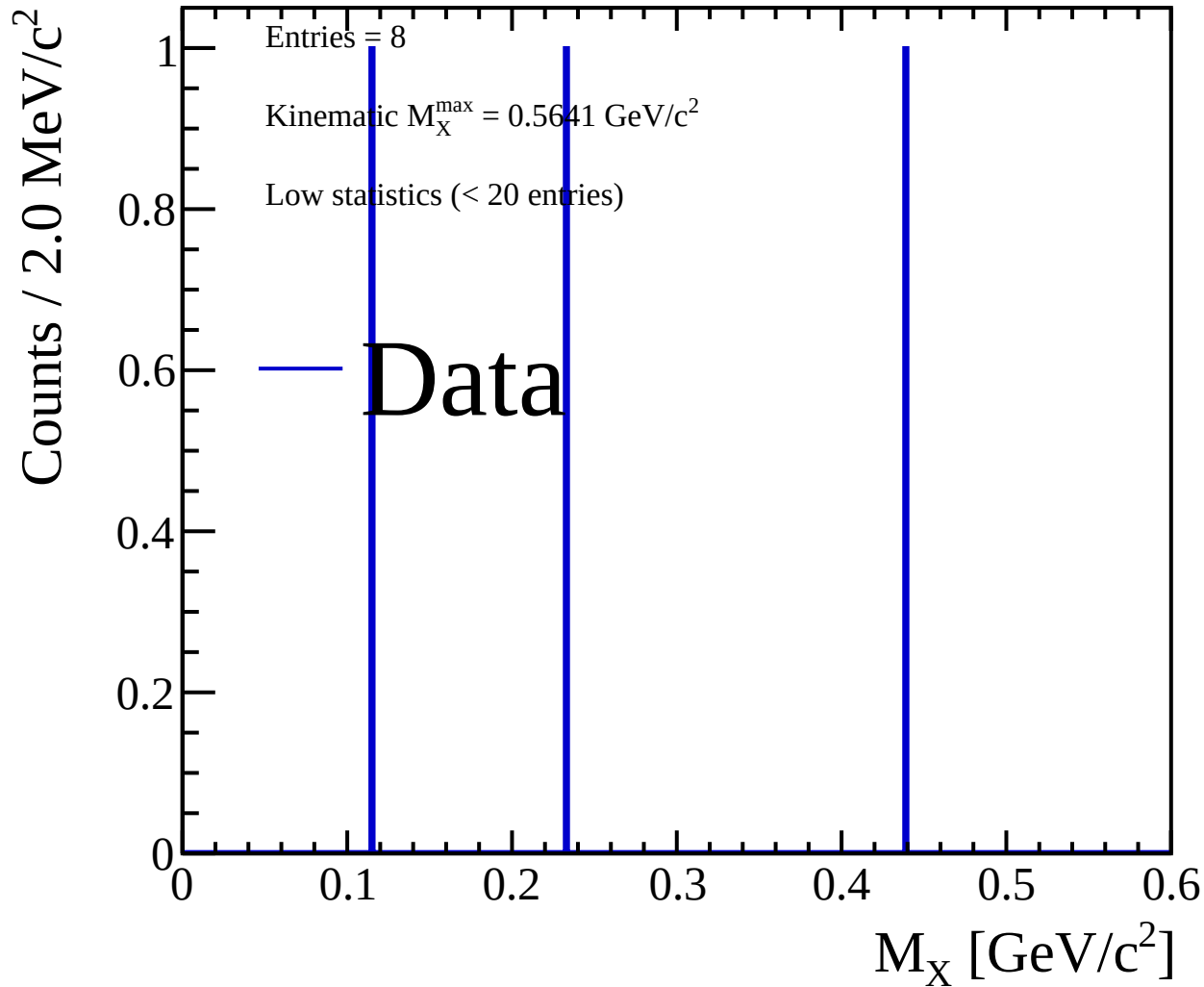
Production-vertex K^- momentum 748--752 MeV/c



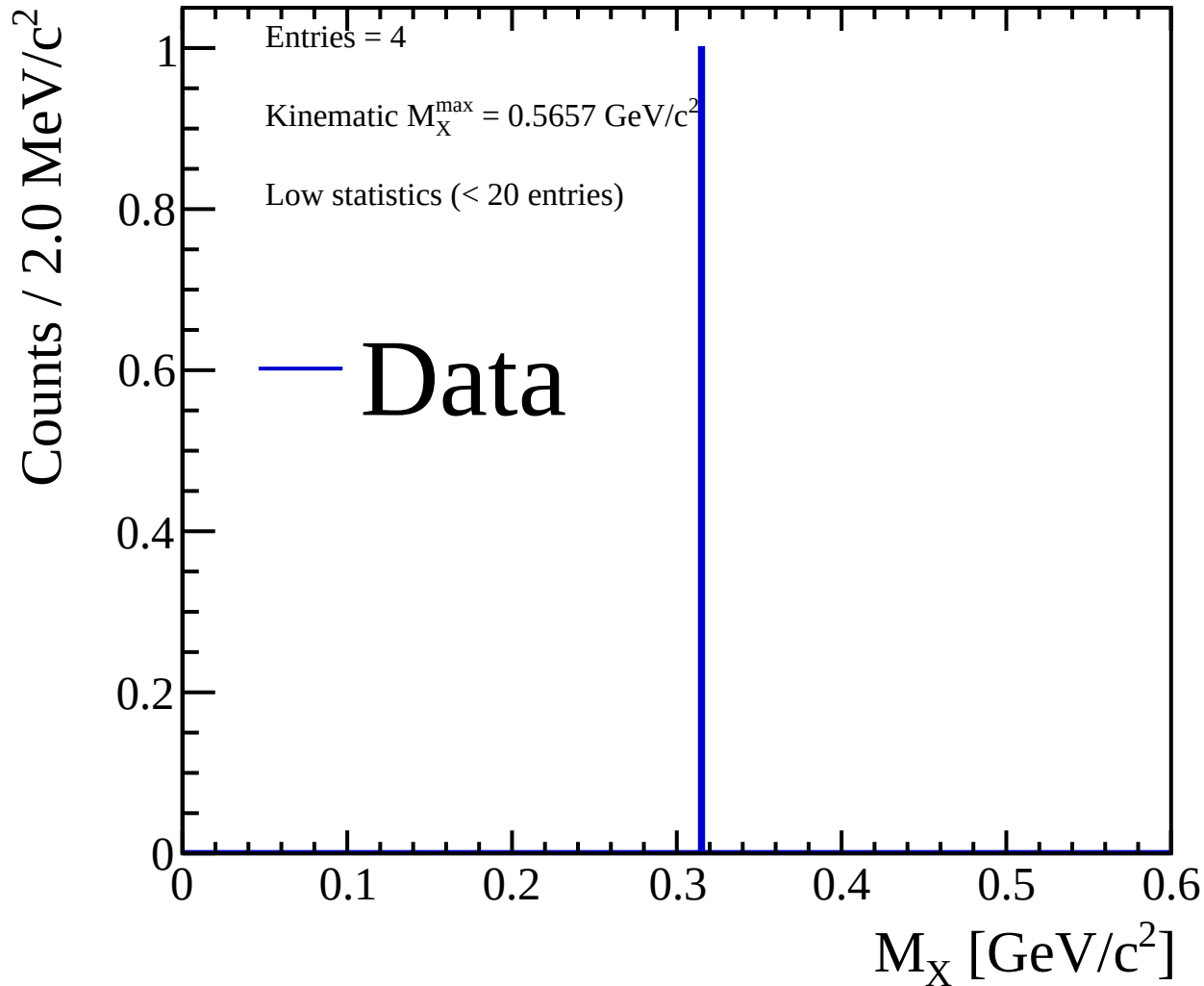
Production-vertex K^- momentum 752--756 MeV/c



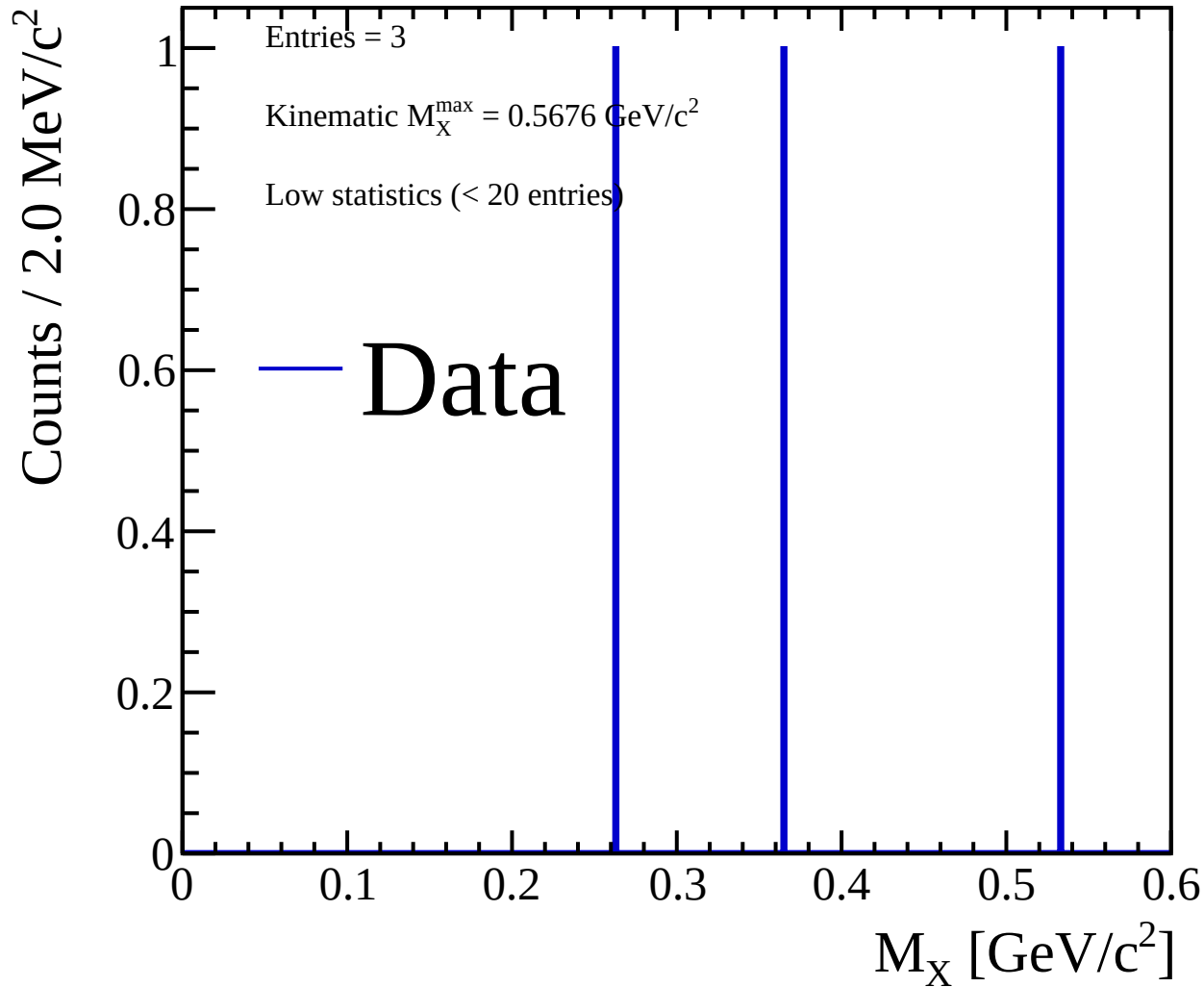
Production-vertex K^- momentum 756--760 MeV/c



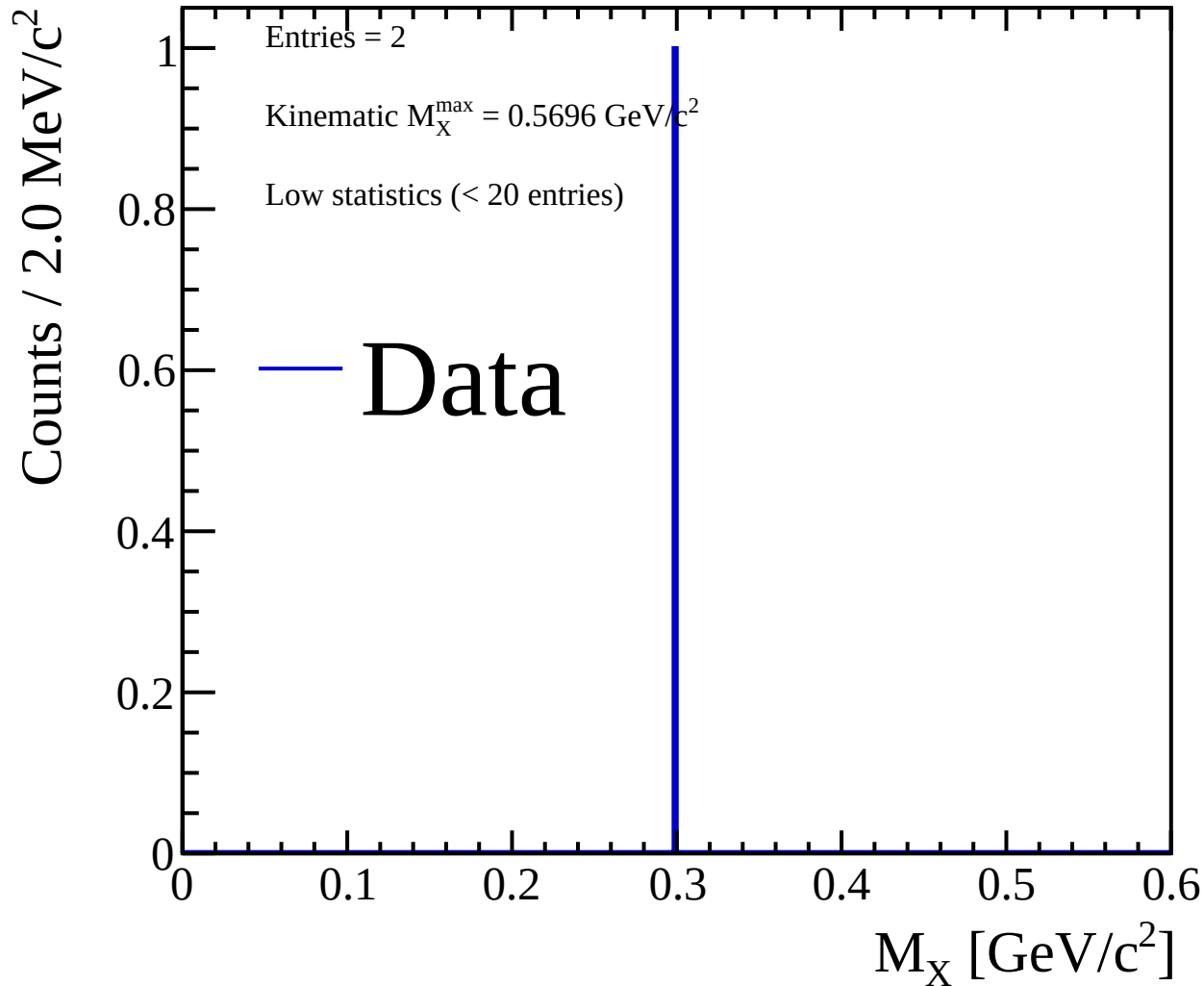
Production-vertex K^- momentum 760--764 MeV/c



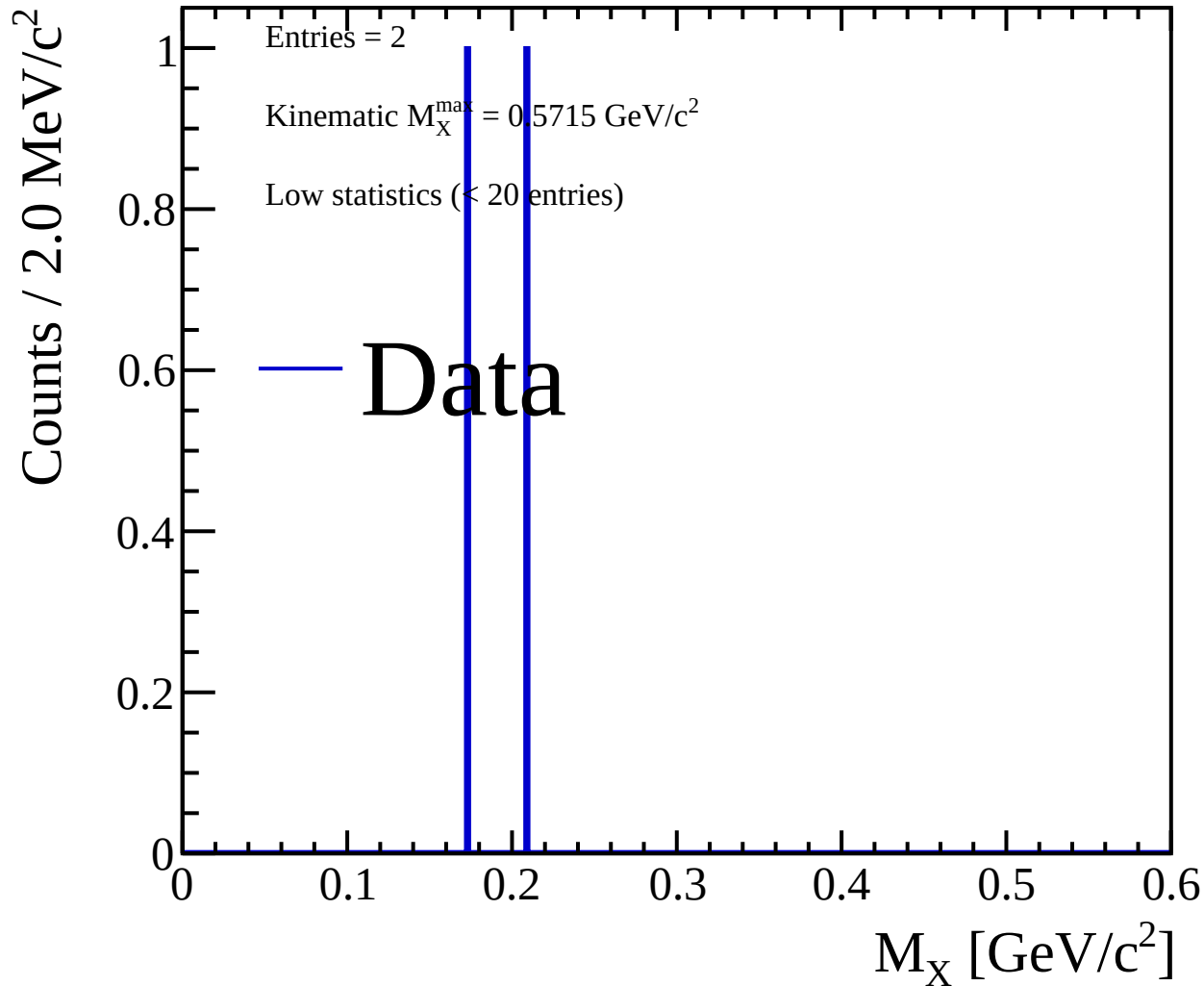
Production-vertex K^- momentum 764--768 MeV/c



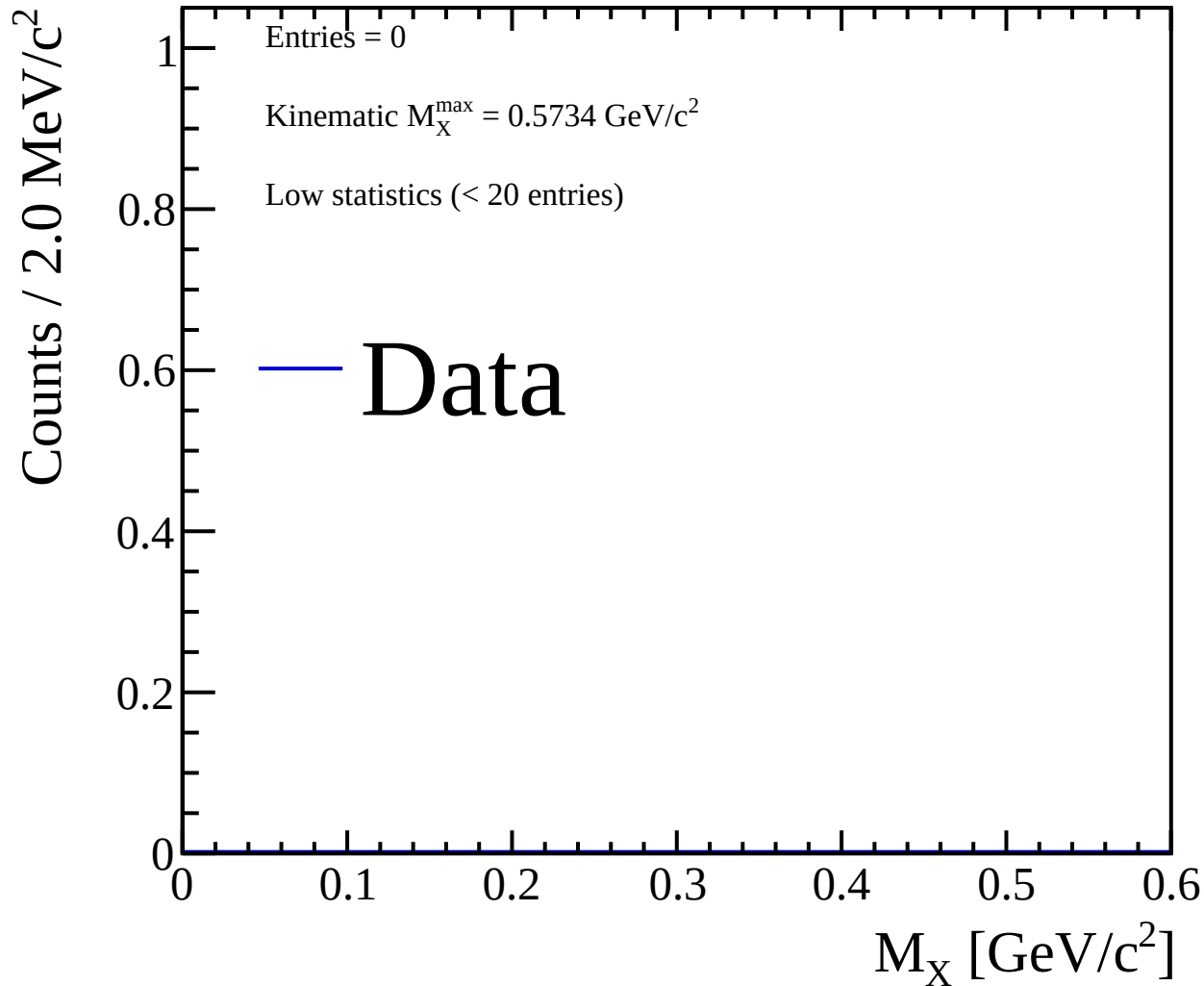
Production-vertex K^- momentum 768--772 MeV/c



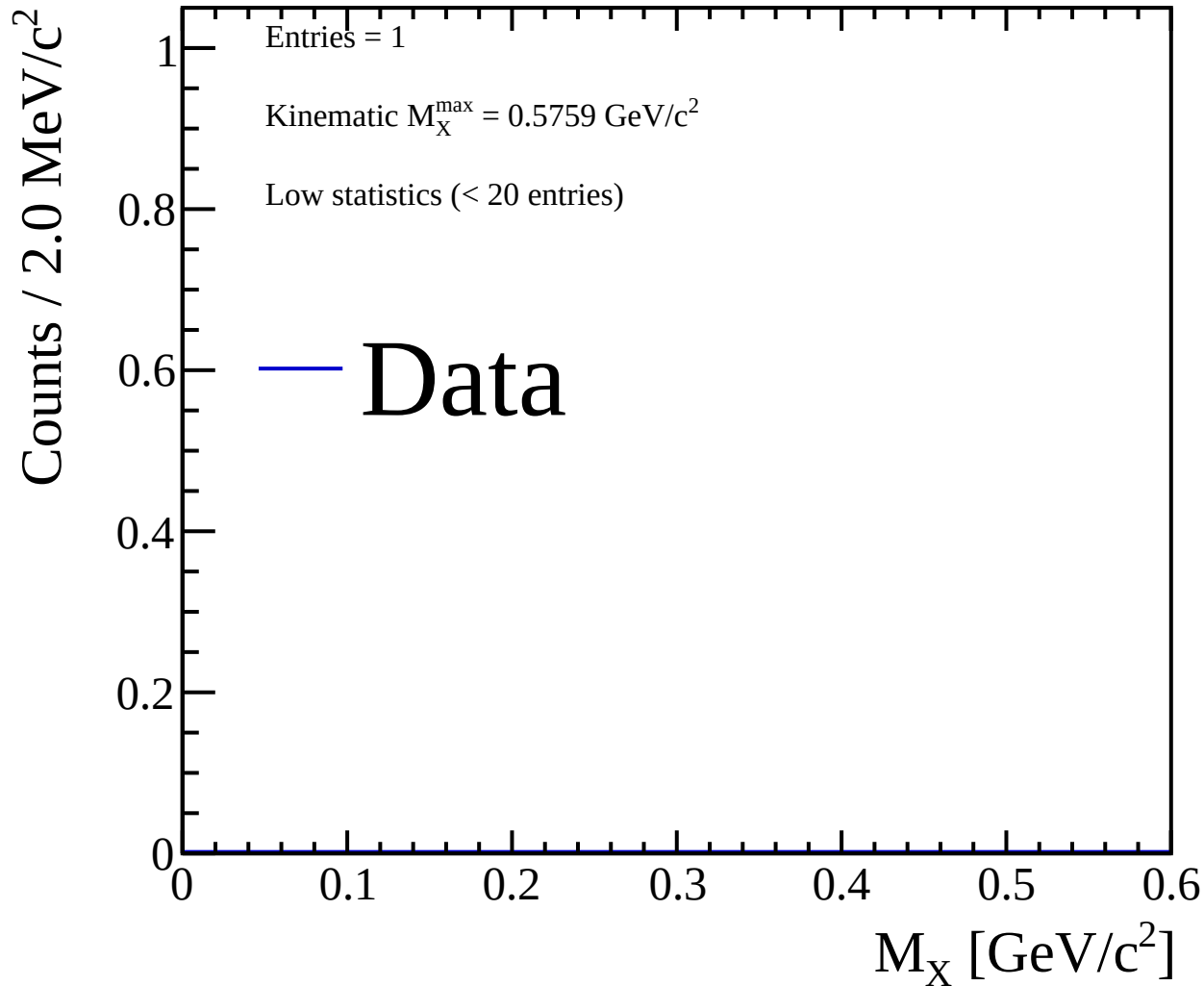
Production-vertex K^- momentum 772--776 MeV/c



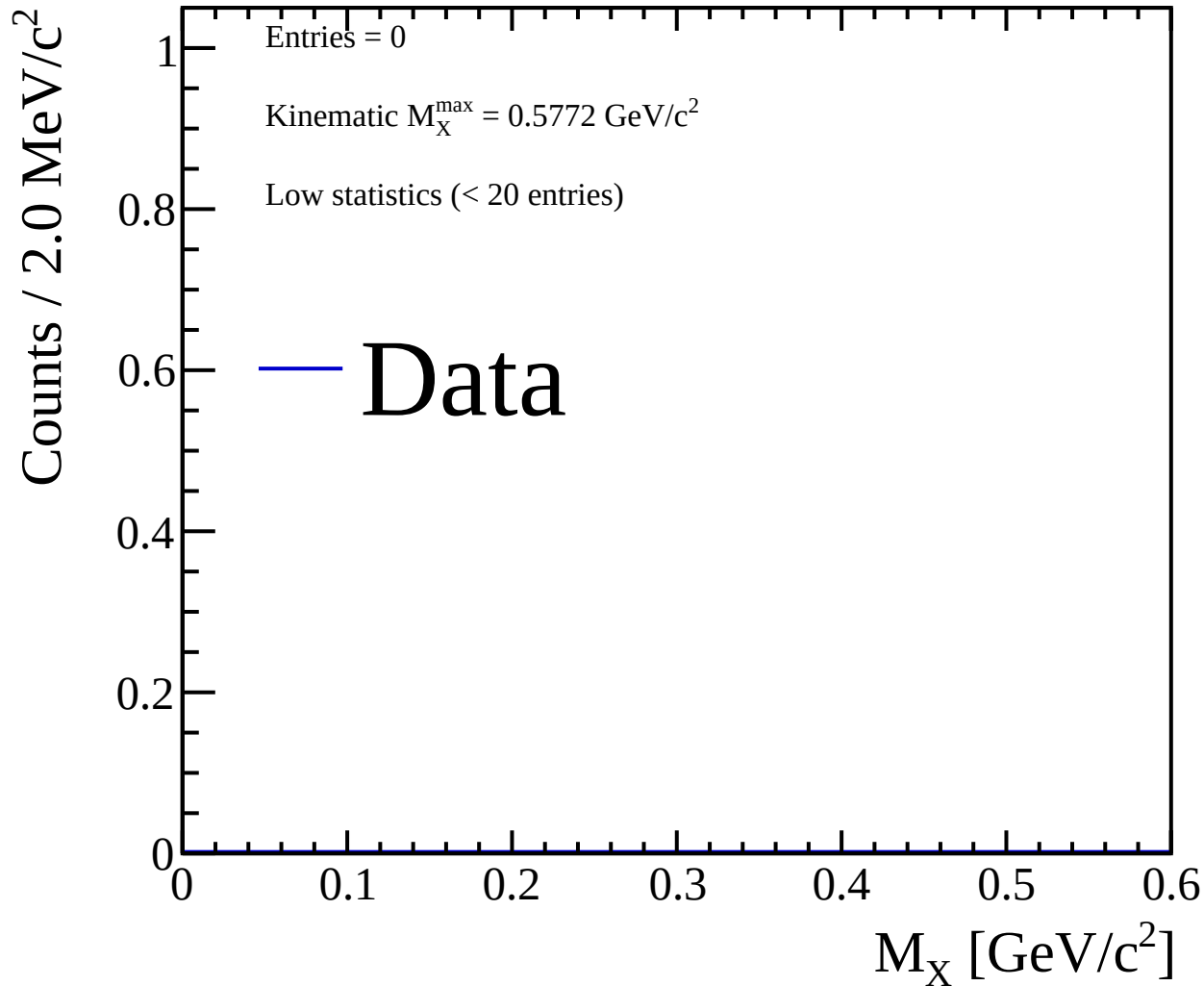
Production-vertex K^- momentum 776--780 MeV/c



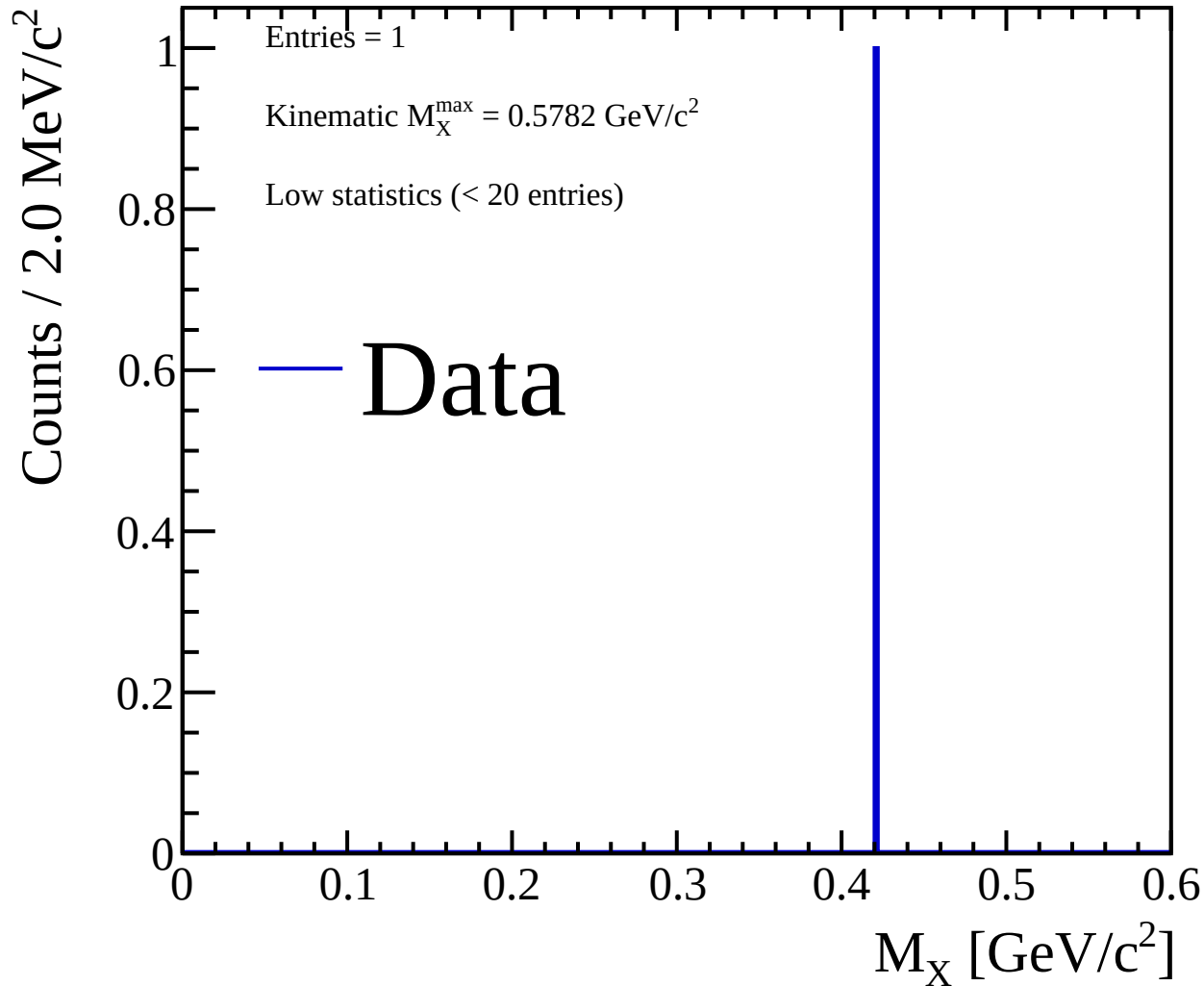
Production-vertex K^- momentum 780--784 MeV/c



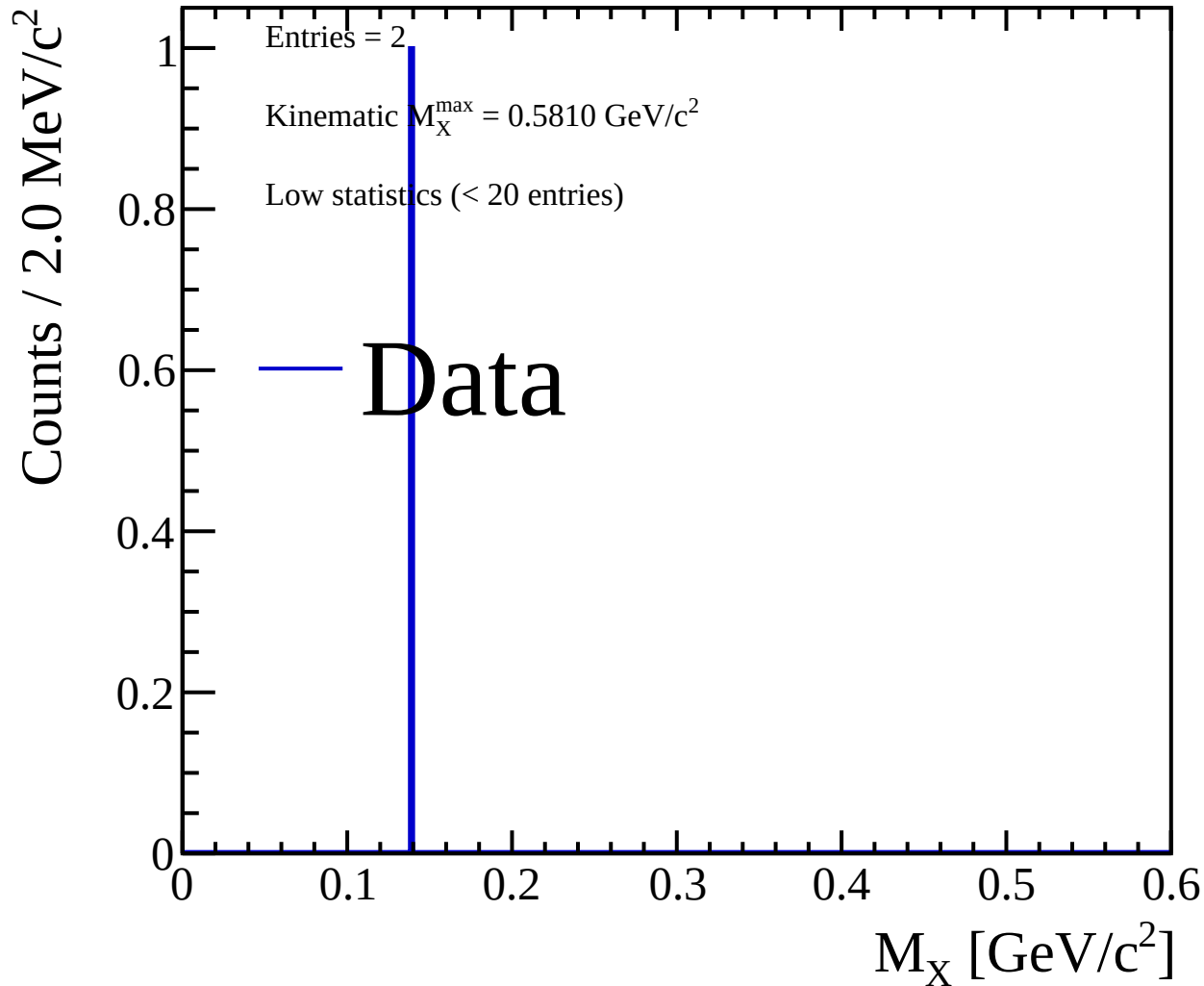
Production-vertex K^- momentum 784--788 MeV/c



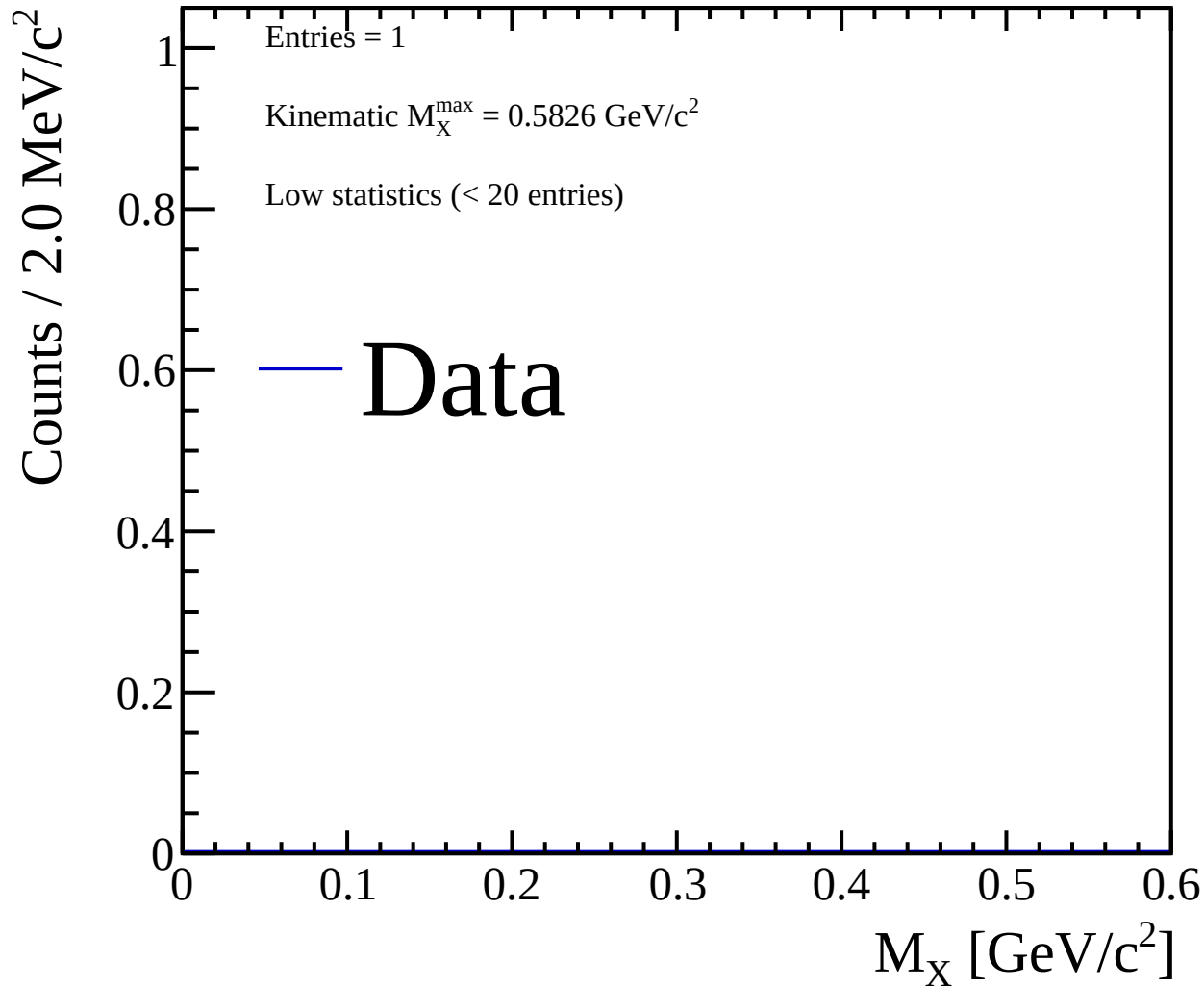
Production-vertex K^- momentum 788--792 MeV/c



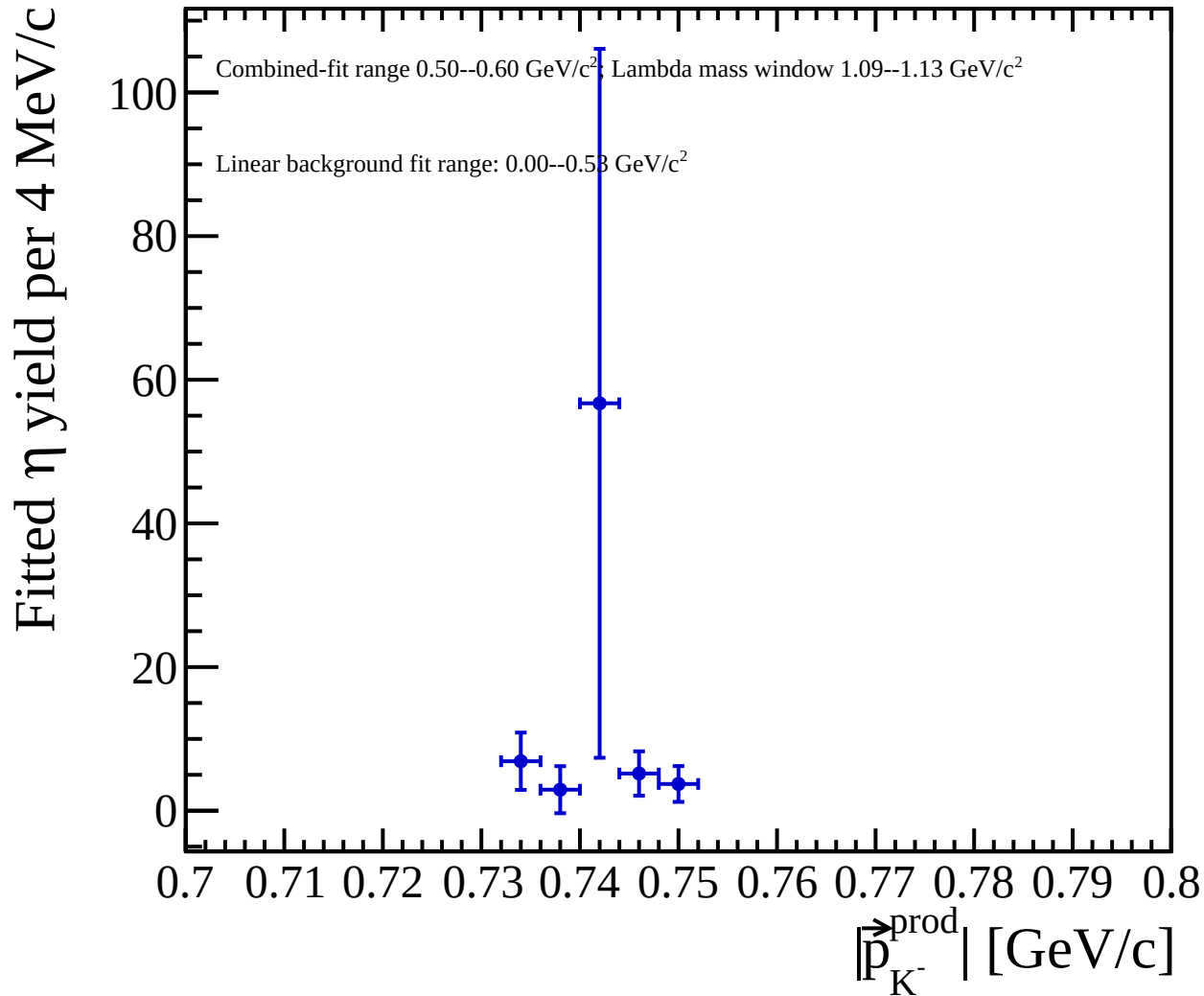
Production-vertex K^- momentum 792--796 MeV/c



Production-vertex K^- momentum 796--800 MeV/c



Gaussian-fit $\Lambda\eta$ yield vs production-vertex K^- momentum



Fitted η peak position vs production-vertex K^- momentum

